

MISI 3: IMPLEMENTASI PRODUKSI

Step 1 : Dashboard Development

```
CREATE VIEW dbo.vw_Budget_Vs_Actual
AS
SELECT
    dw.Tahun,
    dw.Kuartal,
    dw.Bulan,
    duo>Nama_Fakultas,
    duo>Nama_Unit,
    dpa.Grup_Akun,
    dpa>Nama_Akun,
    fa.Nilai_Anggaran,
    -- Realisasi diambil dari kolom Nilai_Kredit di Fact_Transaksi, yang
    biasanya mewakili Pengeluaran
    COALESCE(SUM(ft.Nilai_Kredit), 0) AS Nilai_Realisasi_Transaksi,
    -- Menghitung Varian (Selisih)
    (fa.Nilai_Anggaran - COALESCE(SUM(ft.Nilai_Kredit), 0)) AS
    Nilai_Varian,
    -- Menghitung Persentase Penyerapan/Realisasi
    CASE
        WHEN fa.Nilai_Anggaran > 0 THEN
            (COALESCE(SUM(ft.Nilai_Kredit), 0) / fa.Nilai_Anggaran) * 100
        ELSE NULL
    END AS Persen_Penyerapan
FROM dbo.Fact_Anggaran fa
INNER JOIN dbo.Dim_Waktu dw ON fa.ID_Waktu = dw.ID_Waktu
INNER JOIN dbo.Dim_Unit_Org duo ON fa.ID_Unit = duo.ID_Unit
INNER JOIN dbo.Dim_Pos_Akun dpa ON fa.ID_Akun = dpa.ID_Akun
-- Gabungkan dengan Fact_Transaksi untuk mendapatkan Nilai Realisasi aktual
LEFT JOIN dbo.Fact_Transaksi ft
    ON fa.ID_Waktu = ft.ID_Waktu
    AND fa.ID_Akun = ft.ID_Akun
    AND fa.ID_Unit = ft.ID_Unit
WHERE
    duo.Is_Current_Flag = 'Y' -- Menggunakan nama unit/fakultas yang berlaku
    saat ini
    AND dpa.Is_Current_Flag = 'Y' -- Menggunakan nama akun yang berlaku
    saat ini
GROUP BY
```

```

dw.Tahun, dw.Kuartal, dw.Bulan, duo>Nama_Fakultas, duo>Nama_Unit,
dpa.Grup_Akun, dpa>Nama_Akun, fa.Nilai_Anggaran;
GO

CREATE VIEW dbo.vw_Revenue_And_Receivables_Summary
AS
SELECT
    dw.Tahun,
    dw.Bulan,
    dw.Kuartal,
    dpa.Grup_Akun,
    dpa.Tipe_Akun,
    duo>Nama_Unit,
    -- Total Pendapatan/Penerimaan Kas (Nilai_Debit pada Akun Pendapatan)
    SUM(CASE WHEN dpa.Tipe_Akun = 'Pendapatan' THEN ft.Nilai_Debit
ELSE 0 END) AS Total_Penerimaan_Kas,
    -- Total Piutang (Nilai_Debit pada Akun Piutang)
    SUM(CASE WHEN dpa.Tipe_Akun = 'Piutang' THEN ft.Nilai_Debit ELSE
0 END) AS Total_Piutang_Terbit,
    -- Total Pembayaran Piutang (Nilai_Kredit yang menutup Piutang)
    SUM(CASE WHEN dpa.Tipe_Akun = 'Piutang' THEN ft.Nilai_Kredit ELSE
0 END) AS Total_Pelunasan_Piutang
FROM dbo.Fact_Transaksi ft
INNER JOIN dbo.Dim_Waktu dw ON ft.ID_Waktu = dw.ID_Waktu
INNER JOIN dbo.Dim_Pos_Akun dpa ON ft.ID_Akun = dpa.ID_Akun
INNER JOIN dbo.Dim_Unit_Org duo ON ft.ID_Unit = duo.ID_Unit
WHERE
    dpa.Is_Current_Flag = 'Y'
    AND dpa.Tipe_Akun IN ('Pendapatan', 'Piutang') -- Filter hanya untuk Akun
yang relevan
GROUP BY
    dw.Tahun, dw.Bulan, dw.Kuartal, dpa.Grup_Akun, dpa.Tipe_Akun,
duo>Nama_Unit;
GO

```

```

CREATE VIEW dbo.vw_Top_Vendor_Spending
AS
SELECT
    dw.Tahun,
    dpa.Grup_Akun,
    dpa>Nama_Akun,
    dv>Nama_Vendor,
    dv.Kategori_Vendor,
    dv.Kota,
    -- Total pengeluaran kepada vendor (Nilai_Kredit pada Akun Biaya)

```

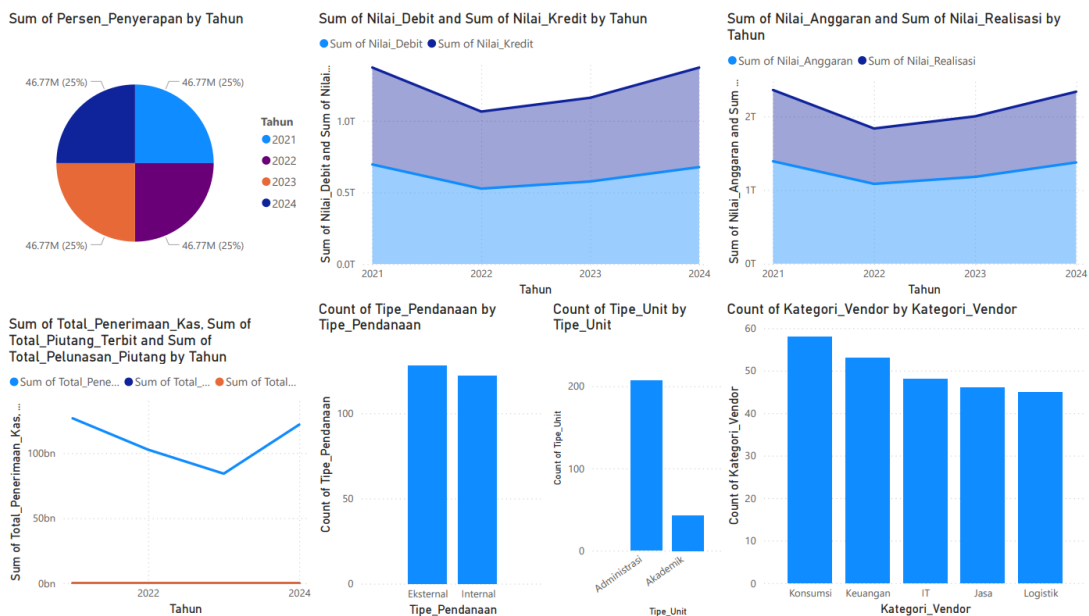
```

SUM(CASE WHEN dpa.Tipe_Akun = 'Biaya' THEN ft.Nilai_Kredit ELSE 0
END) AS Total_Pengeluaran_Ke_Vendor
FROM dbo.Fact_Transaksi ft
INNER JOIN dbo.Dim_Waktu dw ON ft.ID_Waktu = dw.ID_Waktu
INNER JOIN dbo.Dim_Vendor dv ON ft.ID_Vendor = dv.ID_Vendor -- Hanya
menampilkan transaksi yang terhubung ke Vendor
INNER JOIN dbo.Dim_Pos_Akun dpa ON ft.ID_Akun = dpa.ID_Akun
WHERE
    dv.Is_Current_Flag = 'Y' -- Menggunakan nama vendor yang berlaku saat ini
    AND dpa.Is_Current_Flag = 'Y'
    AND dpa.Tipe_Akun = 'Biaya' -- Filter hanya untuk transaksi yang
merupakan biaya/pengeluaran
    AND ft.ID_Vendor IS NOT NULL -- Pastikan hanya transaksi yang
memiliki Vendor
GROUP BY
    dw.Tahun, dpa.Grup_Akun, dpa>Nama_Akun, dv>Nama_Vendor,
dv.Kategori_Vendor, dv.Kota;
GO

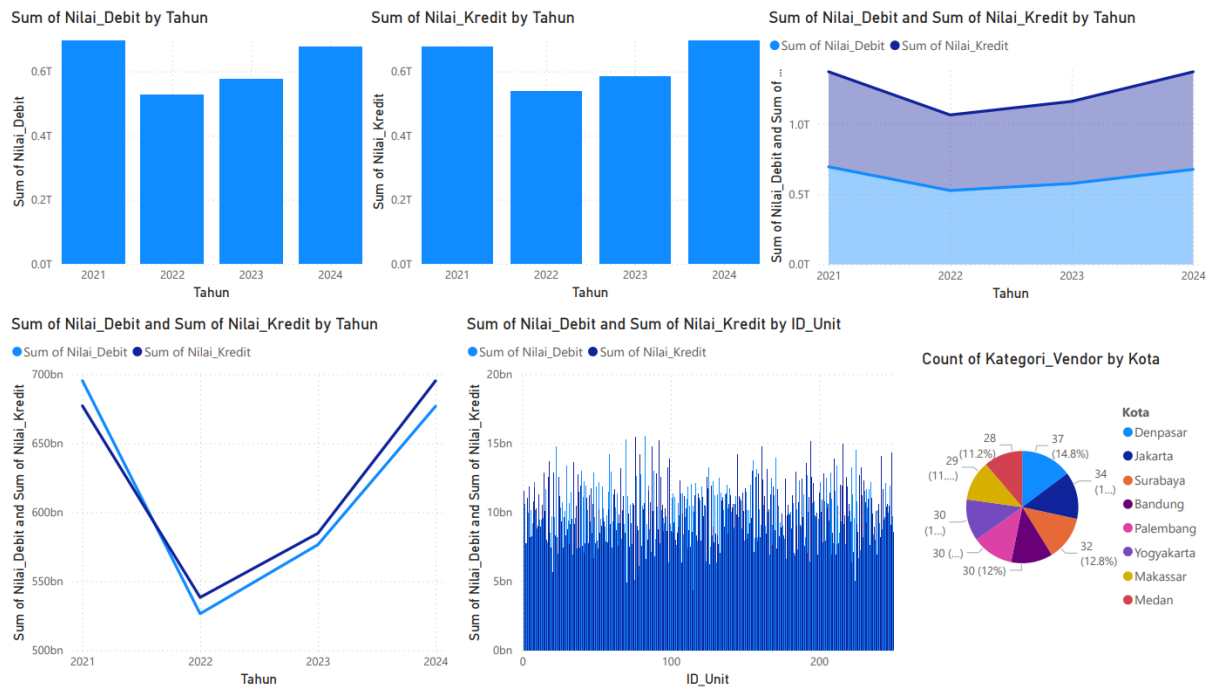
```

Step 2 : Design Power BI

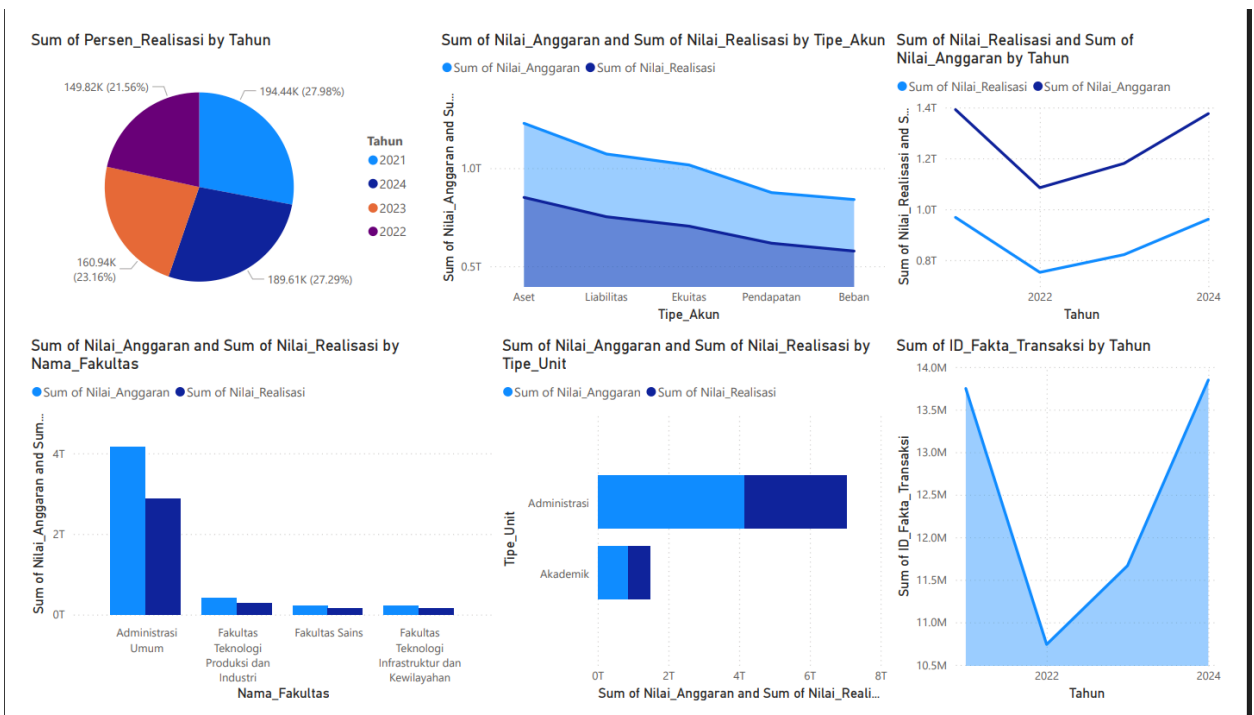
Dashboard 1 : Executive Summary



Dashboard 2: Fakta Transaksi

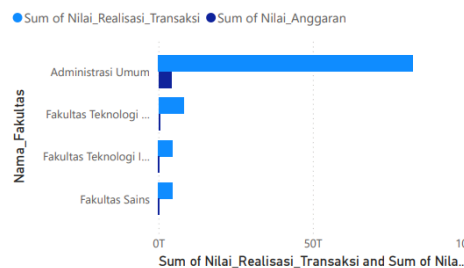


Dashboard 3: Fakta Anggaran

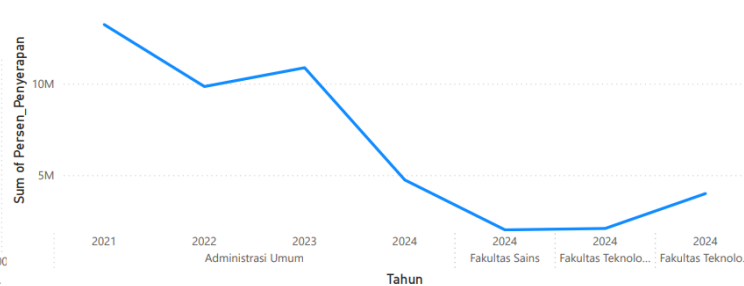


Dashboard 4: Pertanyaan Bisnis

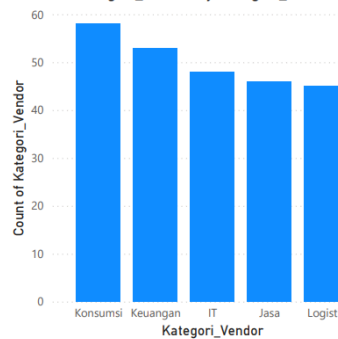
Sum of Nilai_Realisasi_Transaksi and Sum of Nilai_Anggaran by Nama_Fakultas



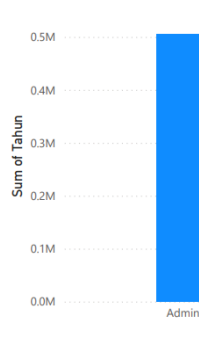
Sum of Persen_Penyerapan by Nama_Fakultas and Tahun



Count of Kategori_Vendor by Kategori_Vendor



Sum of Tahun by Tipe_Unit



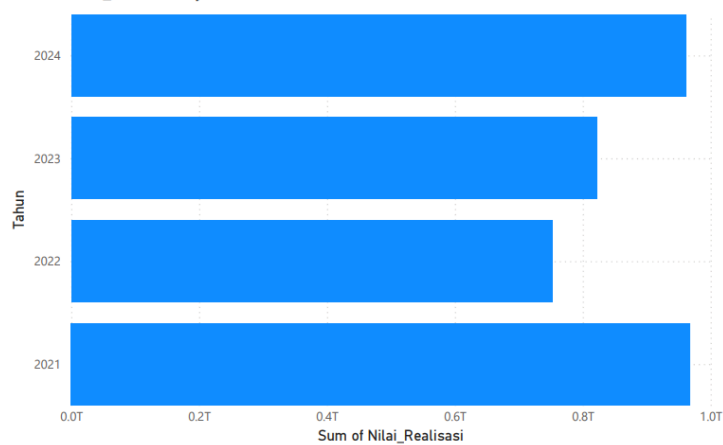
Sum of Total_Piutang_Terbit and Sum of Total_Penerimaan_Kas by Tahun



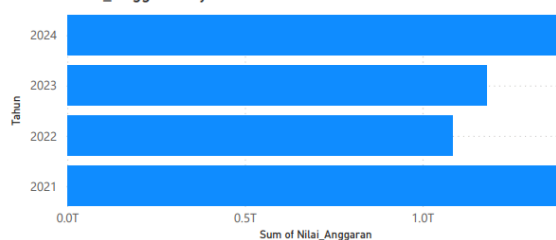
Sum of Persen_Realisasi by Tahun



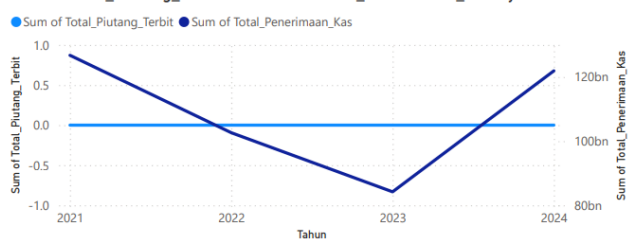
Sum of Nilai_Realisasi by Tahun



Sum of Nilai_Anggaran by Tahun



Sum of Total_Piutang_Terbit and Sum of Total_Penerimaan_Kas by Tahun



Step 3 : Security Implementation

1. Create User Roles

```
USE DM_Keuangan_DW;
GO

-- 1. Create Database Roles
CREATE ROLE db_executive_finance;
CREATE ROLE db_financial_analyst;
CREATE ROLE db_finance_viewer;
CREATE ROLE db_etl_operator;
GO

-- Grant Permissions for Executive Finance
GRANT SELECT ON SCHEMA::dbo TO db_executive_finance; -- Akses baca
ke semua objek (termasuk semua Dim, Fact, dan View)
GRANT EXECUTE ON SCHEMA::dbo TO db_executive_finance; -- Untuk
menjalankan Stored Procedure (misalnya Master ETL)
GO

-- Grant Permissions for Financial Analyst
GRANT SELECT ON SCHEMA::dbo TO db_financial_analyst; -- Akses baca
ke semua Dim, Fact, dan View
-- Asumsi ada Schema STG untuk data staging/penyiapan. Analis mungkin
butuh akses ke sana.
GRANT SELECT, INSERT, UPDATE, DELETE ON SCHEMA::stg TO
db_financial_analyst;
GO

-- Grant Permissions for Finance Viewer (Read-only)
GRANT SELECT ON dbo.vw_Budget_Vs_Actual TO db_finance_viewer;
GRANT SELECT ON dbo.vw_Revenue_And_Receivables_Summary TO
db_finance_viewer;
GRANT SELECT ON dbo.vw_Top_Vendor_Spending TO db_finance_viewer;
-- Berikan akses ke dimensi Waktu dan Unit Organisasi agar mereka dapat
memfilter laporan
GRANT SELECT ON dbo.Dim_Waktu TO db_finance_viewer;
GRANT SELECT ON dbo.Dim_Unit_Org TO db_finance_viewer;
GO

-- Grant Permissions for ETL Operator
GRANT EXECUTE ON SCHEMA::dbo TO db_etl_operator; -- Untuk
```

```
menjalankan Stored Procedure ETL
-- Izin penuh untuk memuat/memperbarui data di semua tabel Dimensi dan
Fakta
GRANT SELECT, INSERT, UPDATE, DELETE ON SCHEMA::dbo TO
db_etl_operator;
GO
```

 Messages

Commands completed successfully.

Completion time: 2025-12-07T13:26:44.1359227+07:00

2. Create Users and Assign Roles

```
--- Create User and Assigns Roles
-- Create SQL Logins
CREATE LOGIN executive_finance_user WITH PASSWORD =
'DW_Keu@ngan_2025';
CREATE LOGIN analyst_finance_user WITH PASSWORD =
'$D!W!_FiNance%89';
CREATE LOGIN viewer_finance_user WITH PASSWORD =
'UnitK3u@ngan#DM';
CREATE LOGIN etl_finance_service WITH PASSWORD =
'Keu@ngan_Dw#ItrA';
GO

USE DM_Kuangan_DW;
GO

-- Create Database Users
CREATE USER executive_finance_user FOR LOGIN executive_finance_user;
CREATE USER analyst_finance_user FOR LOGIN analyst_finance_user;
CREATE USER viewer_finance_user FOR LOGIN viewer_finance_user;
CREATE USER etl_finance_service FOR LOGIN etl_finance_service;
GO

-- Assign to Roles
ALTER ROLE db_executive_finance ADD MEMBER executive_finance_user;
ALTER ROLE db_financial_analyst ADD MEMBER analyst_finance_user;
ALTER ROLE db_finance_viewer ADD MEMBER viewer_finance_user;
ALTER ROLE db_etl_operator ADD MEMBER etl_finance_service;
GO
```

Messages

Commands completed successfully.

Completion time: 2025-12-07T13:35:07.5526883+07:00

3. Implement Data Masking

```
USE DM_Keuangan_DW;
GO

-- Dynamic Data Masking untuk kolom sensitif di Dim_Unit_Org
ALTER TABLE dbo.Dim_Unit_Org
ALTER COLUMN Kode_Unit ADD MASKED WITH (FUNCTION =
'default()');
GO

-- Dynamic Data Masking untuk kolom sensitif di Dim_Vendor
ALTER TABLE dbo.Dim_Vendor
ALTER COLUMN Nama_Vendor ADD MASKED WITH (FUNCTION =
'partial(1,"XX-XX-",2)');
GO

-- Grant UNMASK permission untuk roles spesifik (menggunakan roles yang
sudah kita buat)
GRANT UNMASK TO db_executive_finance;
GRANT UNMASK TO db_financial_analyst;
GO
```

Messages

Commands completed successfully.

Completion time: 2025-12-07T13:36:22.4323077+07:00

4. Implement Audit Trail

```
--- Audit Trail
USE DM_Keuangan_DW;
GO

-- 1. Create Audit Table (DATABASE LEVEL - Sudah benar di
DM_Keuangan_DW)
```



```

CREATE TABLE dbo.AuditLog (
    AuditID BIGINT IDENTITY(1,1) PRIMARY KEY,
    EventTime DATETIME2 DEFAULT SYSDATETIME(),
    UserName NVARCHAR(128) DEFAULT SUSER_SNAME(),
    EventType NVARCHAR(50), -- SELECT, INSERT, UPDATE, DELETE
    SchemaName NVARCHAR(128),
    ObjectName NVARCHAR(128),
    SqlStatement NVARCHAR(MAX),
    RowsAffected INT,
    IPAddress VARCHAR(50),
    ApplicationName NVARCHAR(128) DEFAULT APP_NAME()
);
GO

-- 2. Create Audit Triggers

CREATE TRIGGER trg_Audit_Fact_Transaksi
ON dbo.Fact_Transaksi
AFTER INSERT, UPDATE, DELETE
AS
BEGIN
    SET NOCOUNT ON;
    DECLARE @EventType NVARCHAR(50);
    DECLARE @RowsAffected INT;
    IF EXISTS (SELECT * FROM inserted) AND EXISTS (SELECT * FROM
deleted)
        SET @EventType = 'UPDATE';
    ELSE IF EXISTS (SELECT * FROM inserted)
        SET @EventType = 'INSERT';
    ELSE IF EXISTS (SELECT * FROM deleted)
        SET @EventType = 'DELETE';
    SET @RowsAffected = @@ROWCOUNT;
    INSERT INTO dbo.AuditLog (EventType, SchemaName, ObjectName,
RowsAffected)
    VALUES (@@EventType, 'dbo', 'Fact_Transaksi', @RowsAffected);
END;
GO

CREATE TRIGGER trg_Audit_Fact_Anggaran
ON dbo.Fact_Anggaran
AFTER INSERT, UPDATE, DELETE
AS
BEGIN
    SET NOCOUNT ON;
    DECLARE @EventType NVARCHAR(50);
    DECLARE @RowsAffected INT;

```

```

IF EXISTS (SELECT * FROM inserted) AND EXISTS (SELECT * FROM
deleted)
    SET @EventType = 'UPDATE';
ELSE IF EXISTS (SELECT * FROM inserted)
    SET @EventType = 'INSERT';
ELSE IF EXISTS (SELECT * FROM deleted)
    SET @EventType = 'DELETE';
SET @RowsAffected = @@ROWCOUNT;
INSERT INTO dbo.AuditLog (EventType, SchemaName, ObjectName,
RowsAffected)
VALUES (@EventType, 'dbo', 'Fact_Anggaran', @RowsAffected);
END;
GO

```

```

-- Pindah ke Master Database sebelum membuat Audit Server
USE master;
GO

```

```

-- 3. Enable SQL Server Audit (Server-level) - HARUS DI MASTER
CREATE SERVER AUDIT DataWarehouse_Finance_Audit
TO FILE
(
    FILEPATH = N'D:\LogAudit\'
    ,MAXSIZE = 100 MB
    ,MAX_ROLLOVER_FILES = 10
)
WITH (ON_FAILURE = CONTINUE);
GO

```

```

-- 4. Mengaktifkan Audit Server - HARUS DI MASTER
ALTER SERVER AUDIT DataWarehouse_Finance_Audit WITH (STATE =
ON);
GO

```

```

USE DM_Keuangan_DW;
GO

```

```

-- 5. Create Database Audit Specification - HARUS DI DATABASE TARGET
CREATE DATABASE AUDIT SPECIFICATION
DataWarehouse_Finance_DB_Audit
FOR SERVER AUDIT DataWarehouse_Finance_Audit
ADD (SELECT, INSERT, UPDATE, DELETE ON SCHEMA::dbo BY public);
-- Audit semua operasi CRUD di schema dbo oleh semua pengguna

```

```
GO
```

```
-- 6. Mengaktifkan Spesifikasi Audit Database - HARUS DI DATABASE  
TARGET
```

```
ALTER DATABASE AUDIT SPECIFICATION
```

```
DataWarehouse_Finance_DB_Audit WITH (STATE = ON);
```

```
GO
```

Messages

```
Commands completed successfully.
```

```
Completion time: 2025-12-07T13:42:55.2689329+07:00
```

5. Backup and Recovery Strategy

```
-- Pastikan Database berada dalam Full Recovery Model untuk mengaktifkan  
Log Backup
```

```
USE master;
```

```
ALTER DATABASE DM_Keuangan_DW SET RECOVERY FULL;
```

```
GO
```

```
-----  
-- 1. FULL BACKUP (Cadangan Penuh) - Tanpa Kompresi  
-----
```

```
BACKUP DATABASE DM_Keuangan_DW
```

```
TO DISK = N'D:\BackUp_Keuangan_DW\DM_Keuangan_DW_Full.bak'
```

```
WITH
```

```
INIT, -- Menggantikan isi file backup jika sudah ada
```

```
NAME = N'Full Database Backup - DM_Keuangan_DW',
```

```
STATS = 10; -- Menampilkan progress setiap 10%
```

```
GO
```

```
-----  
-- 2. DIFFERENTIAL BACKUP (Cadangan Diferensial) - Tanpa Kompresi  
-----
```

```
BACKUP DATABASE DM_Keuangan_DW
```

```
TO DISK = N'D:\BackUp_Keuangan_DW\DM_Keuangan_DW_Diff.bak'
```

```
WITH
```

```
DIFFERENTIAL, -- Opsi ini yang membuat backup menjadi Diferensial
INIT, -- Menggantikan isi file backup jika sudah ada
NAME = N'Differential Database Backup - DM_Keuangan_DW',
STATS = 10;
GO
```

```
-----
-- 3. TRANSACTION LOG BACKUP (Cadangan Log Transaksi) - Tanpa
Kompresi
-----
```

```
BACKUP LOG DM_Keuangan_DW
TO DISK = N'D:\BackUp_Keuangan_DW\DM_Keuangan_DW_Log.trn'
WITH
    INIT, -- Menggantikan isi file backup jika sudah ada
    NAME = N'Transaction Log Backup - DM_Keuangan_DW',
    STATS = 10;
GO
```

```
Messages
10 percent processed.
20 percent processed.
30 percent processed.
40 percent processed.
50 percent processed.
60 percent processed.
70 percent processed.
80 percent processed.
90 percent processed.
100 percent processed.
Processed 1928 pages for database 'DM_Keuangan_DW', file 'DM_Keuangan_DW_Data' on file 1.
Processed 2 pages for database 'DM_Keuangan_DW', file 'DM_Keuangan_DW_Log' on file 1.
BACKUP DATABASE successfully processed 1930 pages in 0.160 seconds (94.213 MB/sec).

Completion time: 2025-12-07T19:09:30.6434565+07:00
```

```
Messages
19 percent processed.
38 percent processed.
57 percent processed.
77 percent processed.
96 percent processed.
100 percent processed.
Processed 56 pages for database 'DM_Keuangan_DW', file 'DM_Keuangan_DW_Data' on file 1.
Processed 2 pages for database 'DM_Keuangan_DW', file 'DM_Keuangan_DW_Log' on file 1.
BACKUP DATABASE WITH DIFFERENTIAL successfully processed 58 pages in 0.037 seconds (12.141 MB/sec).

Completion time: 2025-12-07T19:10:41.2770957+07:00
```

```
Messages
37 percent processed.
74 percent processed.
100 percent processed.
Processed 22 pages for database 'DM_Keuangan_DW', file 'DM_Keuangan_DW_Log' on file 1.
BACKUP LOG successfully processed 22 pages in 0.006 seconds (27.994 MB/sec).

Completion time: 2025-12-07T19:11:42.2177968+07:00
```